

A clearer pit wall

Illustrative design concept. The text and values below show how to explain a warning; they are not a live prediction.

SIDEWALL · SILVERSTONE REPLAY

Lap 40 · 20× playback · estimated tyre data

BOX THIS LAP

Front left tyre is at risk

Its safe remaining laps have fallen. Prepare a tyre change at the next pit entry.

Reason: tyre age and worsening pace · Model advice, not a confirmed failure

Which tyre needs attention?

FRONT LEFT

At risk

Health 38 / 100

FRONT RIGHT

Watch

Health 71 / 100

REAR LEFT

OK

Health 84 / 100

REAR RIGHT

OK

Health 86 / 100

What should the team do?

1. **Engineer:** confirm the pit window and tell the driver to box.
2. **Driver:** avoid hard braking until the pit entry.
3. **Crew:** prepare the front left tyre and acknowledge the alert.

Why this recommendation?

Tyre age and recent lap trend raised the model risk. Open details to see the exact temperatures, probability, graph and model source.

Changes to make in the existing app

1. **Put the decision first:** retain the existing call banner, but pair it with one affected tyre, one reason and one next action. Use the existing `call`, `worst_tyre`, `reasons` and `radio` fields.
2. **Translate numbers:** label each tyre OK, Watch or At risk beside its health score. Explain thresholds and avoid implying a score is a physical measurement.
3. **Progressively reveal detail:** default to decision, four tyres and crew action. Put telemetry, two risk charts, life curve, track map and event log behind expandable panels.
4. **Keep evidence visible:** mark replay temperature and pressure as estimated, measured sensor values as measured, and injected debris as what-if. Do not use a single vague source tag for mixed evidence.
5. **Show alert history:** mark when a call began, what changed, whether the crew acknowledged it, and whether the system is connected. This distinguishes a fresh warning from a stale one.

Implementation starting points: `web/pitwall.html` for hierarchy, `web/pitwall.js` for display logic, `web/app.css` for layout, and `sidewall/engine/health.py` for call wording. The current crew view already has acknowledgement and vibration.